MASTER FUNCTIONAL SKILLS (MFS) · WORKED SOLUTIONS ·
MasterFunctionalSkills.co.uk · NCFE Maths Level 1 Specialist
Full Marks (100% Pass) · Verified by Tutor Abdul · Sample 3

ncfe.

Sample Paper: P000350

NCFE Functional Skills Qualification in Mathematics at Level 1 (501/2325/7)

Time Allowed 2 HOURS

You **need** the following to complete this assessment:

- ruler
- calculator

Read each document and activity carefully and attempt to answer **all** activities.

Write your answers in the spaces provided and ensure that your writing is legible.

If extra pages are used, please make sure your name is on them and they are securely fastened to this booklet.

At the end of the assessment hand all documents over to the invigilator as instructed.

DO NOT TURN OVER UNTIL YOU ARE INSTRUCTED TO DO SO BY THE INVIGILATOR.

SAMPLE

For Examiner use only:

Activity number	1	2	3	Total
Total Marks awarded				
Total Marks available	**12**	**16**	**12**	**40**

OFFICIAL MFS MODEL SOLUTION · CANDIDATE & BENCHMARK DETAILS

Learner Name: Master Functional Skills
Learner No: MFS-NCFE-2026
Centre Name: Master Functional Skills (MFS01)
Centre No: MFS01 · Date: 20/10/2026
Signature: Master Functional Skills
Score: Full Marks (100% Pass)
Assessment: Sample 3 · Verified by Tutor Abdul (MasterFunctionalSkills.co.uk)

MASTER FUNCTIONAL SKILLS

ONLINE LEARNING PLATFORM & ACCREDITED COURSES

Pearson Edexcel & Open Awards Accredited Suite · Entry 3, Level 1 & Level 2

Pass Your Functional Skills Exams With Confidence

100% Pass Guarantee | Dyslexia-Friendly Learning | 1-to-1 Expert Tutor Support

Master Functional Skills provides an adaptive, dyslexia-friendly online learning platform tailored for adult learners, apprentices, and healthcare & vocational staff. Everything you need to revise, practice, and pass is at your fingertips.

1. Smart Diagnostic Initial Assessment

Calculate your exact working level and pinpoint knowledge gaps instantly

- **Exact Grade Calculation:** Adaptive assessments calculate your exact level across all topics.
- **Personalised Study Pathway:** Skip what you know; focus 100% on the topics needed to pass.
- **Real-Time Exam Readiness:** Predictive tracker confirms when you are 100% ready to book.

2. Comprehensive Video Lessons & Worked Solutions

Bite-sized visual explainers covering every single Maths & English curriculum standard

- **Step-by-Step Video Lessons:** Master percentages, algebra, perimeter, ratio, and reverse fractions.
- **Multi-Ink Handwriting Models:** Visual 3-colour methods showing working, carries, and examiner tips.
- **Tutor Shortcuts & Golden Rules:** Proven non-calculator mental arithmetic hacks and BIDMAS tricks.

3. Full Mock Exams & Real-Time Performance Analytics

Practice in exam conditions and watch your scores and confidence climb

- **Authentic Exam Papers:** Timed Section A & Section B papers formatted exactly like Pearson Edexcel.
- **Instant Topic Breakdown:** View percentage scores across Number, Measure/Shape, and Data.
- **Historical Score Tracking:** Monitor past attempts to ensure continuous improvement before exam day.

Ready to Master Your Functional Skills & Pass First Time?

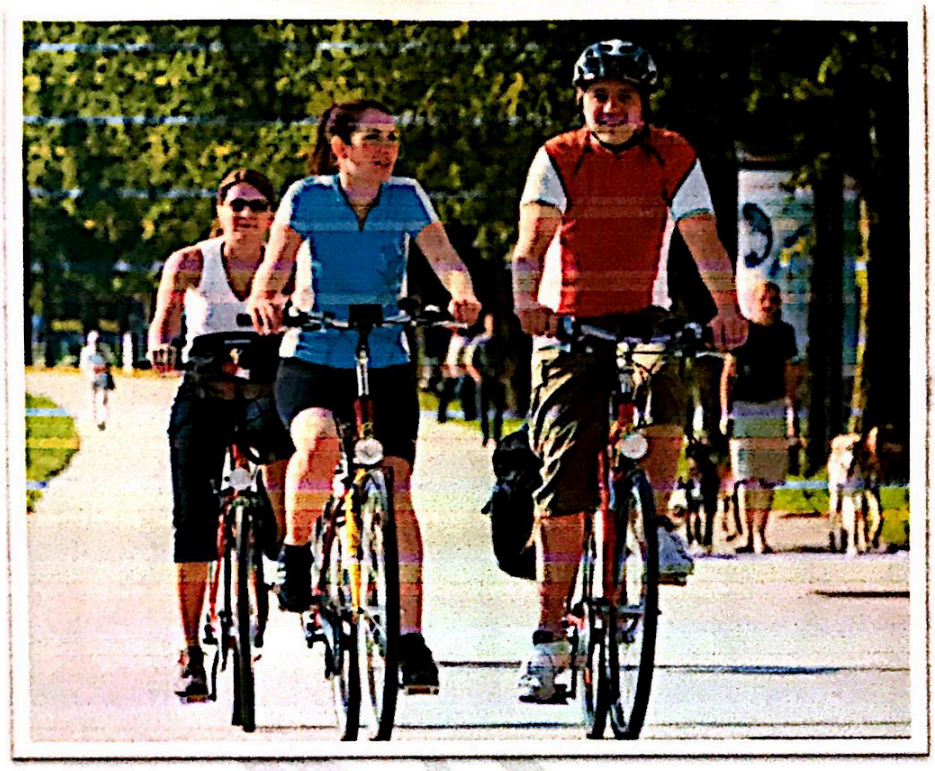
Join thousands of healthcare staff, apprentices, and adult learners across the UK.

Instant 24/7 access on desktop, tablet, and mobile with no software to install.

START LEARNING TODAY AT

MasterFunctionalSkills.co.uk

Cycling



This assessment is about cycling.

Complete activities 1, 2 and 3 based on the documents provided for each activity.

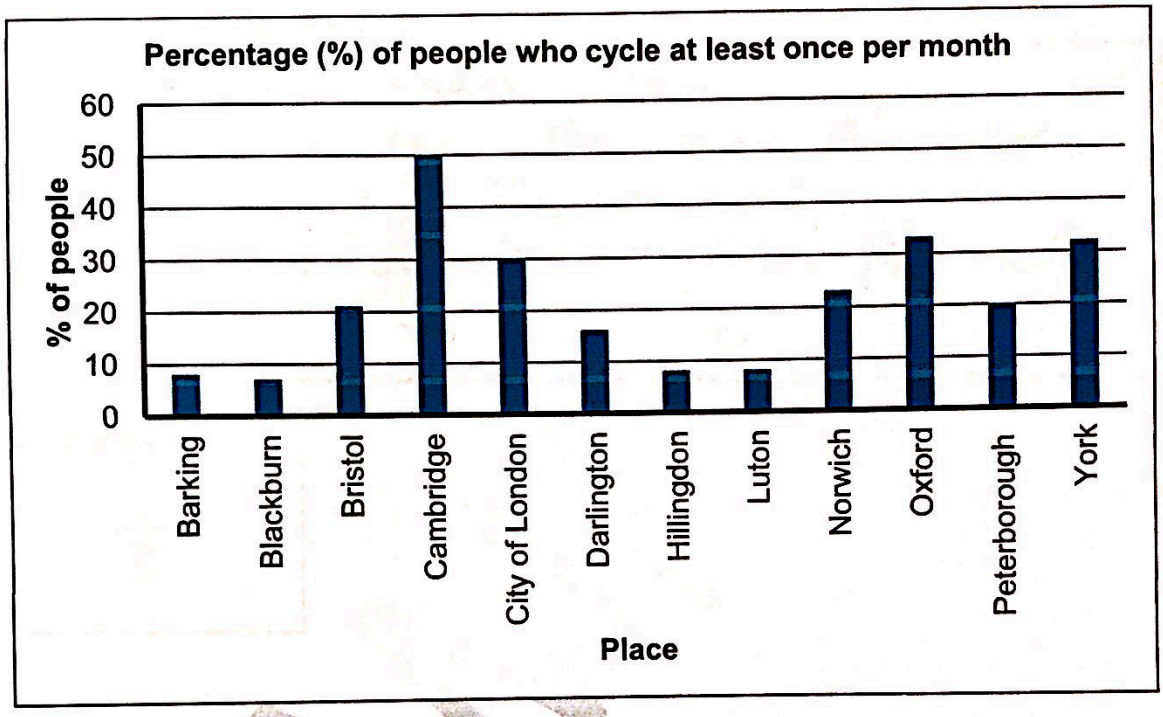
CamScanner

Activity 1

Task A

Adults and children cycle to work and school. They also cycle for fitness and fun.

The chart below shows the percentage of people who cycle at least once per month in different places.



1. Which place has more than twice the percentage of people who cycle in Peterborough?

Marks available: 2

You must show your working:

Peterborough = 20%.
20% × 2 = 40%.
Only Cambridge is more than 40%.

Your answer:

Cambridge

CamScanner

2. Find the places where more than a quarter of the people cycle at least once per month.

Show the number of these places as a fraction of all the places in the chart. Show your answer in its **simplest form**.

Marks available: 3

You must show your working:

A quarter means 25%.
Places more than 25% are Cambridge, City of London, Oxford, York.
4 places out of 12 total places.
$\frac{4}{12} = \frac{1}{3}$

Your answer:

$\frac{1}{3}$

SAMPLE

CamScanner

Task B

1. In a town near London a teacher asks pupils in one class how they travel to school.

The table shows the results. There are 35 in the class.

Method of travel	Number of pupils
Car	15
Bus	3
Cycle	7
Walk	10

What is the ratio of pupils who walk compared to the pupils who travel by car? Show your answer in its **simplest form**.

Marks available: 2

You must show your working:

Walk = 10
Car = 15

10 : 15
= 2 : 3

Your answer:

2 : 3

SAMPLE

CamScanner

2. What is the percentage of pupils in the class who cycle to school?

Marks available: 3

You must show your working:

7 cycle out of 35 in the class

$\frac{7}{35} = \frac{1}{5} = 20\%$

Your answer:

20%

3. In the town, 35% of journeys made to college are by car. 25% walk or cycle. The rest travel by bus.

What is the likelihood that a journey is made by bus? Show your answer as a **decimal.**

Marks available: 2

You must show your working:

100 - 35 - 25 = 40%
which is 0.4.

Your answer:

0.4

Total marks available: 12

SAMPLE

CamScanner

Activity 2

Jay, Vin and Sam are keen cyclists.

Task A

1. Jay measures the length of a cycling route using a mobile phone app. The app shows the distance as 31560 metres (m).

What is this distance in kilometres (km)?

Marks available: 1

Space for your working:

31560 ÷ 1000 = 31.56km

Your answer:

31.56km

SAMPLE

CamScanner

2. Vin completes laps of a cycle track with the times shown in the table. Each lap is 250 metres (m).

Lap number	Time (seconds)
Lap 1	32.14
Lap 2	31.94
Lap 3	31.68
Lap 4	31.89
Lap 5	32.14
Lap 6	32.71
Lap 7	32.57

What is the range of the lap times?

Marks available: 2

You must show your working:

32.71 − 31.68 = 1.03 seconds.

Your answer:

1.035

SAMPLE

CamScanner

Task B

1. Sam cycles for 2 hours and 30 minutes each week. She cycles on 5 days of the week and for the same length of time each day.

How many minutes does she cycle each day?

Marks available: 2

You must show your working:

2 hours 30 minutes = 150 minutes
150 ÷ 5 = 30 minutes

Your answer:

30 mins

2. Sam uses 720 calories in 60 minutes of cycling. She drinks a can of lemonade which contains 168 calories.

How many minutes does she need to cycle to use the 168 calories?

Marks available: 3

You must show your working:

$\frac{720}{60}$ = 12 calories per minute.
168 ÷ 12 = 14 minutes.

Your answer:

14 mins

SAMPLE

CamScanner

3. Jay cycles and goes to the gym. Each week $\frac{2}{3}$ of his exercise is cycling and the rest is in the gym.

What **percentage** of his exercise is in the gym?

Marks available: 2

You must show your working:

$1 - \frac{2}{3} = \frac{1}{3}$

$\frac{1}{3}$ is 33.3%

Your answer:

33.3%

SAMPLE

CamScanner

Task C

> 'Cycling is one of the easiest ways to fit exercise into your daily routine because it's also a form of transport. It saves you money, gets you fit and is good for the environment.'
>
> *- NHS (2015, http://www.nhs.uk/Livewell/fitness/Pages/Cycling.aspx)*

1. Rita is thinking about buying a bicycle to travel to college.

The costs to buy and maintain a good bicycle are:

- New bicycle £300
- Cycling helmet £40
- Reflective high visibility cycling belt £15
- Servicing £50 per year
- Insurance £25 per year

She expects to keep the bicycle for 2 years. What is the total cost over **2 years**?

Marks available: 2

You must show your working:

Upfront costs £300 + £40 + £15 = £355
Yearly costs £50 + £25 = £75
Total = £355 + 2 × £75 = £505

Your answer:

£505

SAMPLE

CamScanner

2. Rita has 2 options: to buy and maintain the bicycle or join her local gym which costs £30 per month.

How much money will she save over 2 years by cycling instead of using the gym?

Marks available: 4

You must show your working:

Cycling costs £505
Gym costs £30 × 12 = £720
£720 − £505 = £215

Your answer:

£215

Show how you can check your answer:

£215 + £505 = £720

Total marks available: 16

SAMPLE

CamScanner

Activity 3

Task A

Jay completes a journey of 3.8 kilometres (km). The journey takes 15 minutes.

The average speed in km per hour is equal to the distance in km divided by the time in hours.

What is the average speed for the journey in km per hour?

Marks available: 4

You must show your working:

15 mins = $\frac{1}{4}$ hour

Speed = $3.8 \div \frac{1}{4} = 15.2$ km/h

Your answer:

15.2 km/h

Show how you can check your answer:

$15.2 \times \frac{1}{4} = 3.8$

SAMPLE

CamScanner

Task B

1. Rita carried out a survey of people who cycled to college from Monday to Friday.

The number of cyclists per day is shown in the table.

Day	Monday	Tuesday	Wednesday	Thursday	Friday
Number	87	122	136	142	128

What is the average (mean) number of cyclists per day?

Marks available: 2

You must show your working:

5 days $\frac{87+122+136+142+128}{5} = 123$

Your answer:

123

SAMPLE

CamScanner

2. Rita recorded the number of cyclists on a Saturday for 4 weeks.

The results for the first 3 weeks are shown in the table.

The average (mean) number of cyclists on a Saturday is 43.

Complete the table with the missing number at Week 4.

Week	Week 1	Week 2	Week 3	Week 4
Number	38	42	39	53

Your answer

Marks available: 3

Use the space below for your working:

4 days (38 + 42 + 39 + Week 4) ÷ 4 = 43

38 + 42 + 39 + Week 4 = 43 × 4

38 + 42 + 39 + Week 4 = 172

119 + Week 4 = 172

Week 4 = 172 − 119

Week 4 = 53

SAMPLE

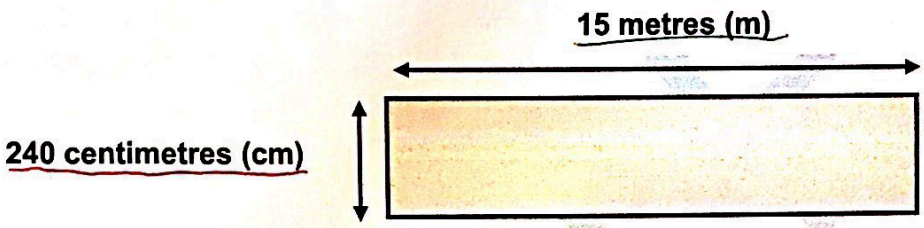
CamScanner

Task C

The college is adding a roof to cover the bicycle racks.

The diagram below shows the measurements of each roof.

Not to scale



What is the total area of roof needed for each shelter? Show your answer with the **correct units.**

Marks available: 3

You must show your working:

240cm = 2.4m
Area = 2.4 × 15 = 36 m²

Your answer:

36 m²

Total marks available: 12

End of assessment

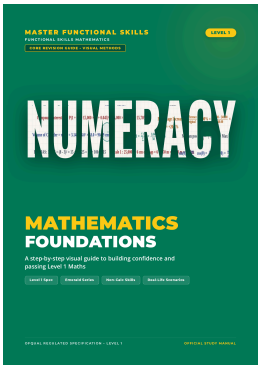
SAMPLE
CamScanner

MASTER FUNCTIONAL SKILLS STORE

OFFICIAL REVISION GUIDES, WORKBOOKS & FLASHCARDS

Written by accredited specialist tutors to match the exact 2026 Pearson Edexcel standards

Everything You Need To Revise, Practice & Pass Your Maths Level 1 Exam



Core Curriculum Manual

Maths Level 1
Core Study Guide

- Full official curriculum coverage
- Section A Non-Calc & Sec B Calc
- Worked models & golden rules
- Dyslexia-friendly visual format

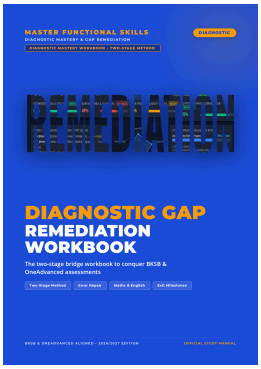
Spiral Bound Print & PDF

1,000+ Practice Drills

Maths Level 1
Companion Workbook

- 1,000+ exam-style questions
- Progressive difficulty drills
- Full worked step-by-step key
- Targeted fix for weak areas

Spiral Bound Print & PDF



Realistic Mock Papers

Maths Level 1
Official Mock Packs

- Full mock exam paper sets
- Authentic Pearson Edexcel style
- Detailed mark schemes included
- Grade boundaries & pass marks

Exam Booklet Pack & PDF

Pocket Flashcards A6

Maths Level 1
Pocket Revision Cards

- Compact A6 cards for quick study
- Essential formulas & conversions
- Dual-sided question & answer
- High-yield mental arithmetic rules

Boxed Card Deck & Print PDF

Order Official Study Guides & Revision Cards Online

Free Next-Day UK Tracked Shipping · Instant PDF Downloads Included

Trusted by training providers, apprentices & adult learners nationwide.

VISIT THE ONLINE STORE

MasterFunctionalSkills.co.uk